3	C	Take direction to the right and upwards as positive. $s_x = u_x t$ $t = \frac{s_x}{u_x} = \frac{3.4\cos 30°}{5.0\cos 60°}$ $v_y = u_y + a_y t = 5.0\sin 60° + (-9.81)\left(\frac{3.4\cos 30°}{5.0\cos 60°}\right) = -7.2240 \text{ m s}^{-1}$ $\tan\beta = \frac{v_y}{v_x}$, where β is the angle that the final velocity makes with the horizontal $\beta = \tan^{-1}\left(\frac{7.2240}{5.0\cos 60°}\right) = 70.911°$ $\theta = 70.911° - 30° = 40.911° = 41°$ OR $v_y{}^2 = u_y{}^2 + 2a_y s_y$ $v_y = \pm\left[(5.0\sin 60°)^2 + 2(-9.81)(-3.4\sin 30)\right]^{½} = -7.2183 \text{ m s}^{-1}$ or 7.2183 m s^{-1} (reject) $\tan\beta = \frac{v_y}{v_x}$, where β is the angle that the final velocity makes with the horizontal $\beta = \tan^{-1}\left(\frac{7.2183}{5.0\cos 60°}\right) = 70.897°$ $\theta = 70.897° - 30° = 40.897° = 41°$

4	**A**	Based on the given graph, the direction upwards is positive.